

Project 10: Pre-Capstone Preparation

Title

SOC and Digital Forensics Investigation System – Pre-Capstone Preparation

Objective

The objective of this project is to prepare the datasets, tools, workflows, and investigation environment required for the final capstone project. This preparation ensures that all forensic and SOC analysis components are properly organized and ready for investigation.

Tools Used

- Kali Linux
- VirtualBox
- Wireshark
- Autopsy
- Volatility Framework
- Linux Log Utilities
- ChatGPT for reporting assistance

Folder Structure Prepared

Capstone_Project/

- Logs/
- Network/

- Disk/
- Memory/
- Screenshots/
- Reports/

Prepared Datasets and Evidence

Log Analysis

Authentication and failed login logs were organized for future SIEM and security analysis.

Network Analysis

Wireshark screenshots and captured network traffic evidence were prepared for investigation.

Disk Forensics

Autopsy investigation screenshots and deleted file evidence were organized.

Memory Analysis

Volatility Framework outputs and memory analysis screenshots were prepared.

Investigation Workflow

Log Collection



SIEM Analysis



Network Analysis



Disk Investigation



Memory Analysis



Incident Report Generation

Conclusion

The required forensic datasets, workflows, tools, and evidence files were successfully prepared for the final capstone project. The environment is now ready for SOC monitoring, forensic analysis, and incident investigation activities.